## I - Before simulation

**① Definition of the simulation parameters dictionaries :**
- global_data
- grid_data
- p_reference_data
- test_data
- resampling_data

**② $(R, \gamma)$ resampling grids creation**

Run functions in : creation_grilles_RG.py

Inputs : global_data, grid_data

Output : grilles_R_G (folder)

**③ Initialisation step :**
- truth particle trajectory creation
- generation of $n$ random particles and associated trajectories

Run functions in :
- creation_fichier_particule_reference.py

Inputs : global_data, grid_data, p_reference_data, current_time=0

Output : sauvegarde_trajectoire_ref.json stored in data_initialisation

- creation_fichier_particule_joblib.py

Inputs : global_data, grid_data, test_data, current_time=0

Output : $n$ folders particule_n stored in data_initialisation

These functions are based on the functions in file trajectoire_particules.py

## II - Simulation of propagation

**④ Data assimilation algorithm - Particles propagation**

Run : main.py

Inputs : global_data, grid_data, p_reference_data, test_data, resampling_data

Output :

- step x particle folders step_x stored in output_data_and_figures
- step x selection folders step_x stored in 'dossier_output' = output_test

main call the fonction propagation_step_function

propagation_step_function function successively calls the functions from files
- trajectoire_particules.py
- trajectoire_pas_i.py
- type_selection.py
- trajectoire_particules_reechanti.py
- deplacement_pas_i.py
- algorithme_reechantillonnage.py
- calcul_vraisemblance_poids.py

## III - Simulation results

**⑤ Particle evolution over steps**

- particle position over steps
- particle parameters distribution over steps

Run :
- plot_particles_positions.py and genere_animation_gif.py

Inputs : output_test, global_data, p_reference_data, test_data

Output : displays stored in directory_images

- plot_distribution_values.py

Inputs : global_data, grid_data, p_reference_data, test_data

Output : displays stored in display_boxplots

## Parameters dictionaries

① **global_data** : global parameters of the situation
'step time' (s), 'p_load' (caldera unloading, Pa), 'radius load' (caldera radius, m), 'step vectors' (discretisation for display, m), 'nb free propag' ($n\times$ 'step time' = time of free propagation), 'grid RG nb' (number of resampling $(R, \gamma)$ possibilities)

② **grid_data** : grid discretisation
'xmin' (m), 'xmax' (m), 'zmin' (m), 'zmax' (m), 'discretisation step' (m)

③ **p_reference_data** : particle truth parameters
'x0_ref' 'z0_ref' (Coordinates of source location, m), 'R_ref', 'G_ref', 'mu_ref' (viscosity, Pa.s), 'E_ref' (rigidity, Pa), 'V_ref' (volume of magma, m$^3$, 'Step max' (surface-reaching step)

④ **test_data** : Simulation parameters
'Nb particles' (number of particles), 'Assim. window' (number of steps per assimilation window), 'Selection type' (systematic / stratified / multinomial / residual), 'Observation type' (regulier / normal / random), 'Observation step (regular case)' (regular distance between 2 observed points), 'Nb observations (other cases)' (Number of available observations, 'dossier output' (results folder for results)

⑤ **test_data** : ranges of resampling Okada parameters and other parameters
'rd_xc_min','rd_xc_max', 'rd_zc_min','rd_zc_max' (source centroïd) ; 'rd_lg_min', 'rd_lg_max' (lenght) ; 'rd_open_min' 'rd_open_max' (opening) ; 'rd_dip_min', 'rd_dip_max' (dip) ; 'rd_veloc_min','rd_veloc_max' (velocity) ; 'percentage' (couple $(R, \gamma)$ choice)